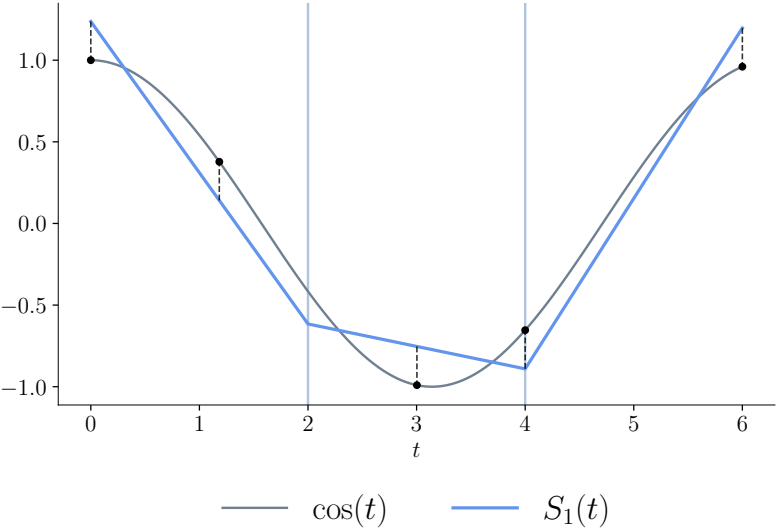
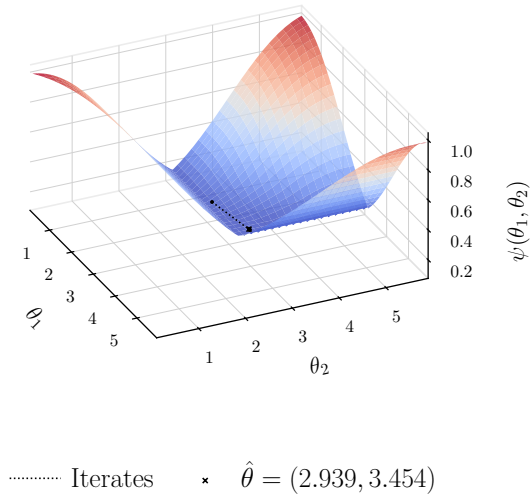


cos k=2 m=1

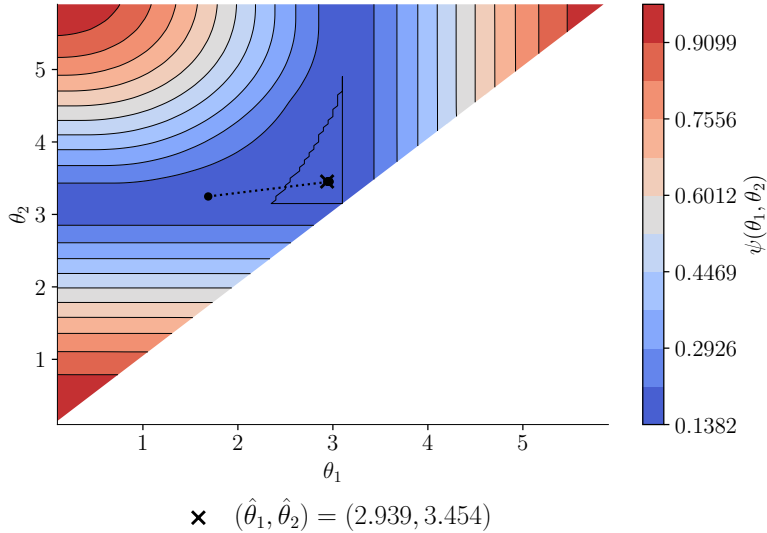
Initial approximation



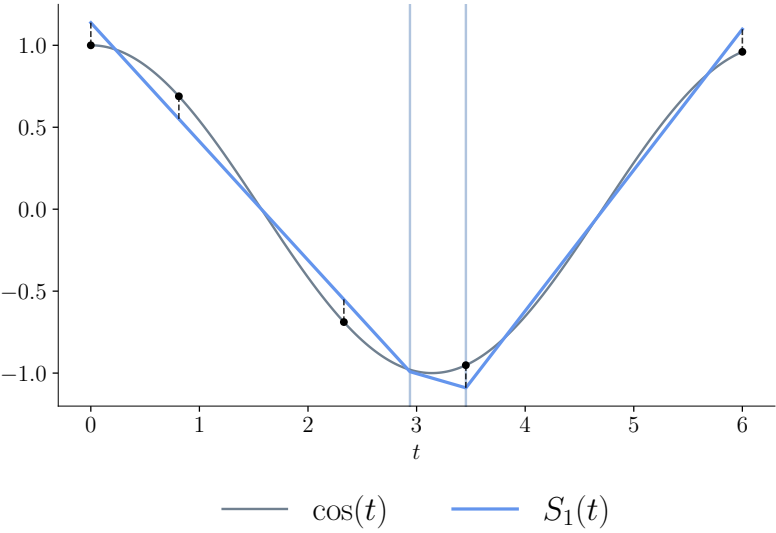
Objective surface and descent path



Contour plot and descent path



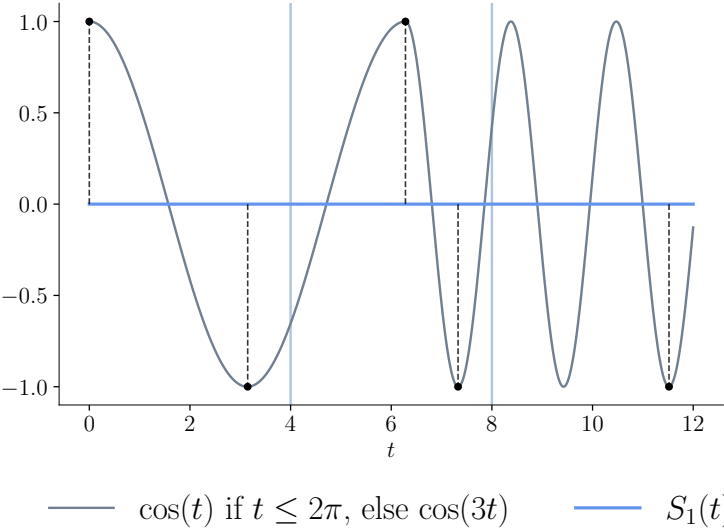
Final approximation



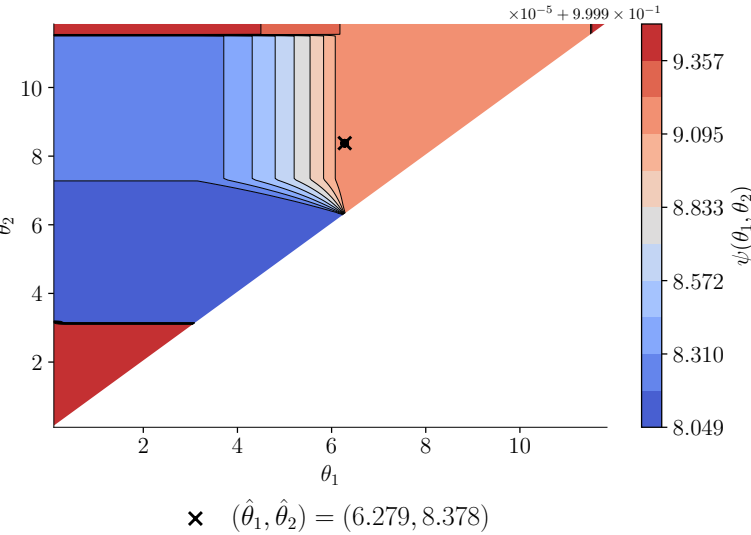
Initial max deviation: 0.236942
Initial knots: [2. 4.]
Nürnberg start: [1.6875 3.24783325]
Theta opt: [2.93888492 3.45407933]
Final max deviation: 0.138529
Iterations: 9

cos_if_else k=2 m=1

Initial approximation

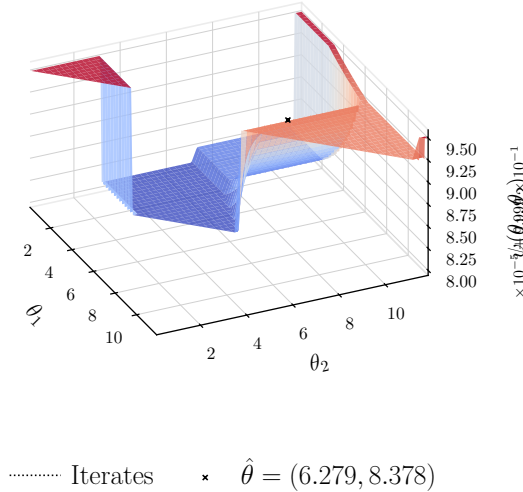


Contour plot and descent path

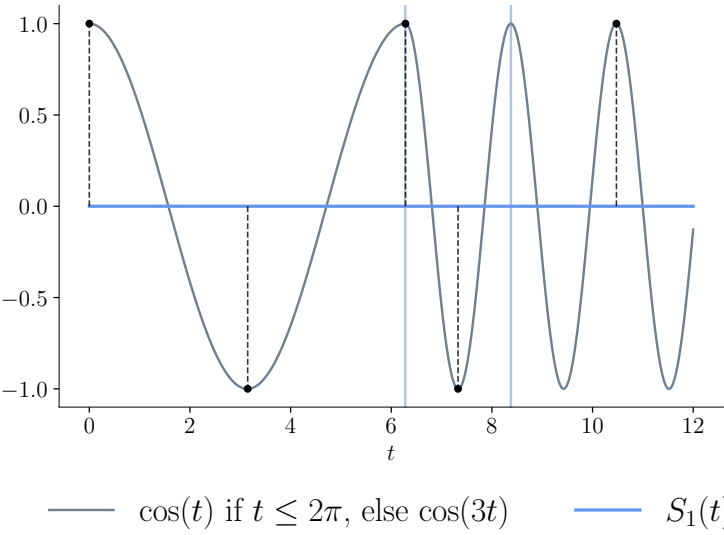


Initial max deviation: 0.999984
Initial knots: [4. 8.]
Nürnberg start: [6.27927928 8.37802217]
Theta opt: [6.27925996 8.37802259]
Final max deviation: -1.00003
Iterations: 9

Objective surface and descent path

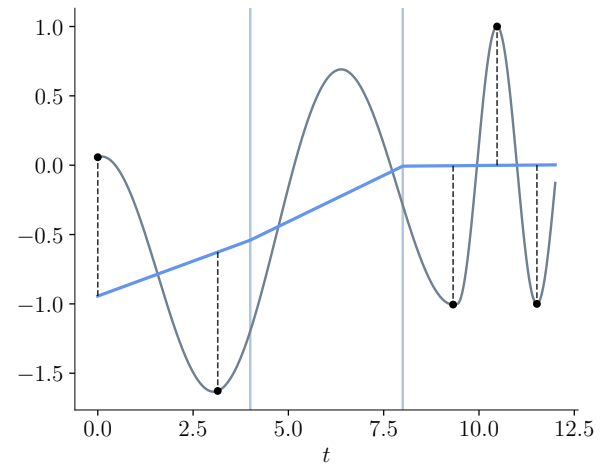


Final approximation



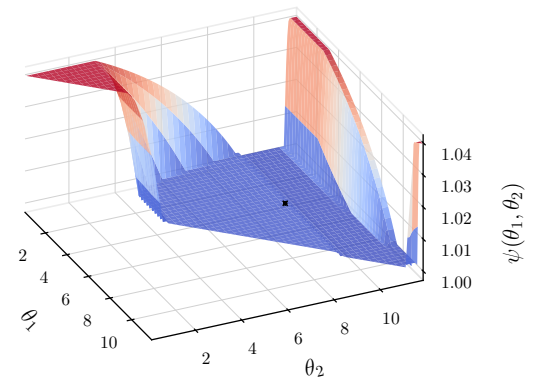
cos_weird k=2 m=1

Initial approximation



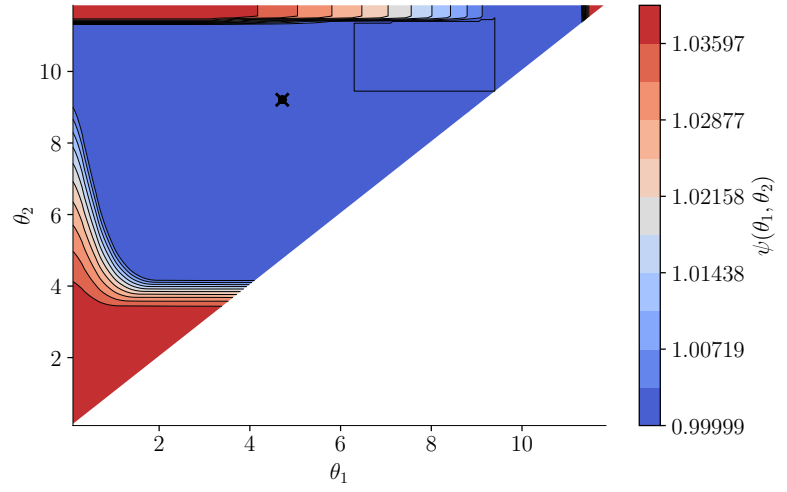
— $\cos(t) + 0.1(t - 3\pi)$ if $t \leq 3\pi$, else $\cos(3t)$ — $S_1(t)$

Objective surface and descent path



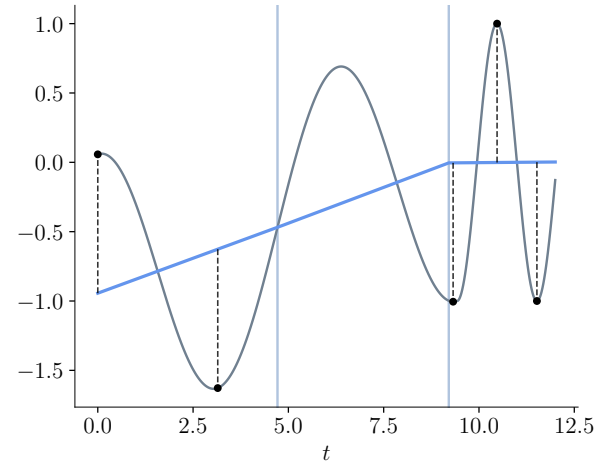
..... Iterates × $\hat{\theta} = (4.714, 9.208)$

Contour plot and descent path



× $(\hat{\theta}_1, \hat{\theta}_2) = (4.714, 9.208)$

Final approximation

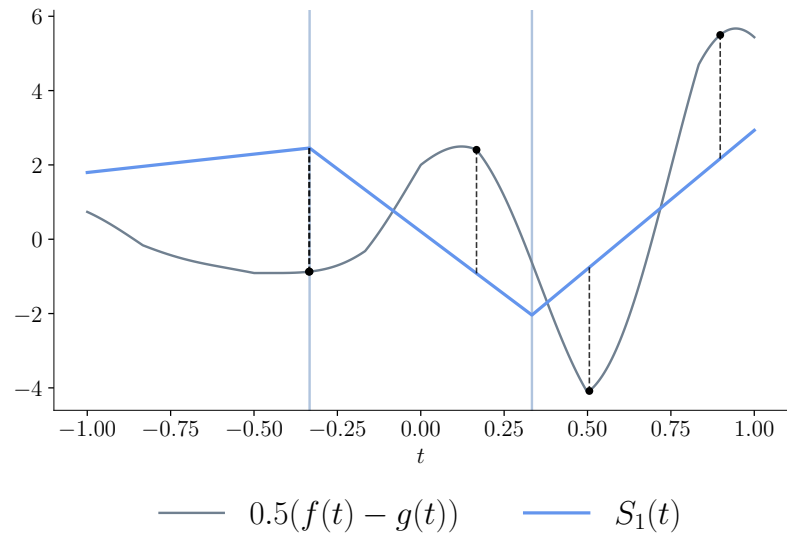


— $\cos(t) + 0.1(t - 3\pi)$ if $t \leq 3\pi$, else $\cos(3t)$ — $S_1(t)$

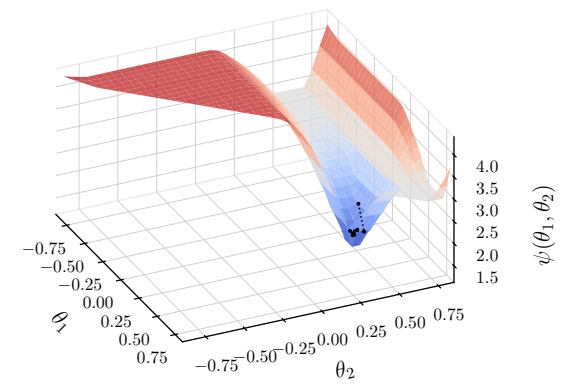
Initial max deviation: 1.00117
 Initial knots: [4. 8.]
 Nürnberg start: [4.71240234 9.20787988]
 Theta opt: [4.71430778 9.2079298]
 Final max deviation: 1.00121
 Iterations: 9

f_g k=2 m=1

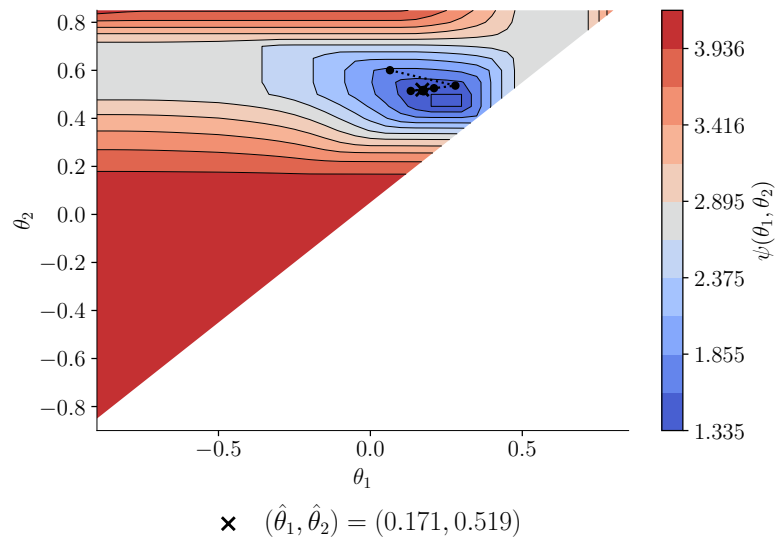
Initial approximation



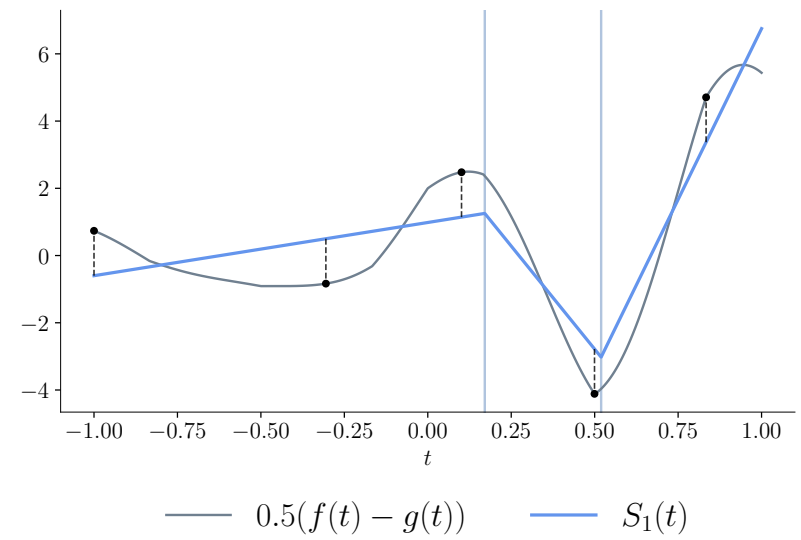
Objective surface and descent path



Contour plot and descent path



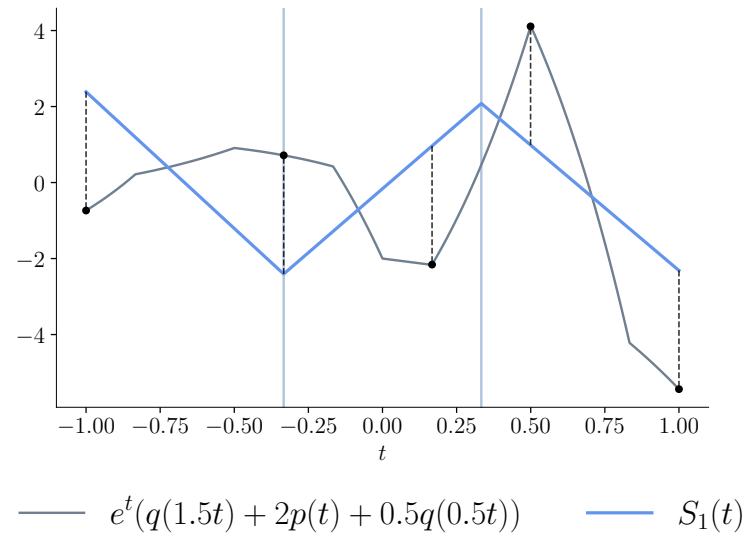
Final approximation



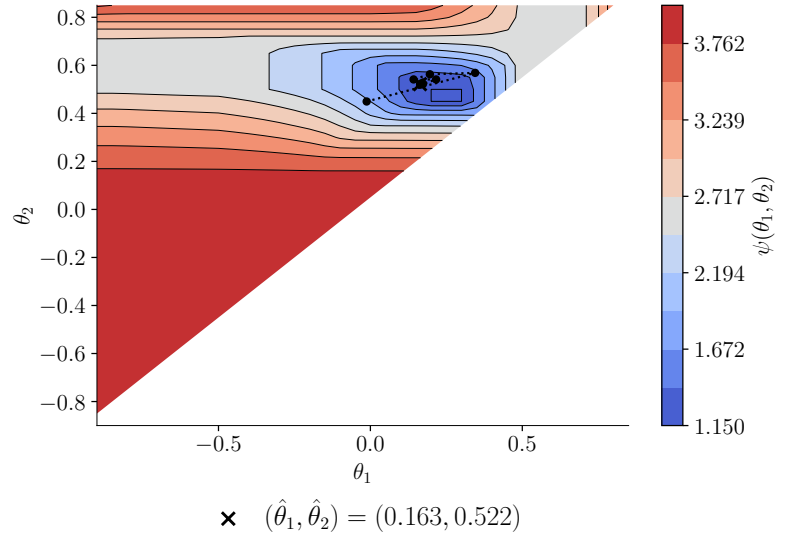
Initial max deviation: 3.32654
 Initial knots: [-0.33333333 0.33333333]
 Nürnberg start: [0.06445312 0.60053393]
 Theta opt: [0.17077252 0.51938792]
 Final max deviation: 1.33634
 Iterations: 9

g k=2 m=1

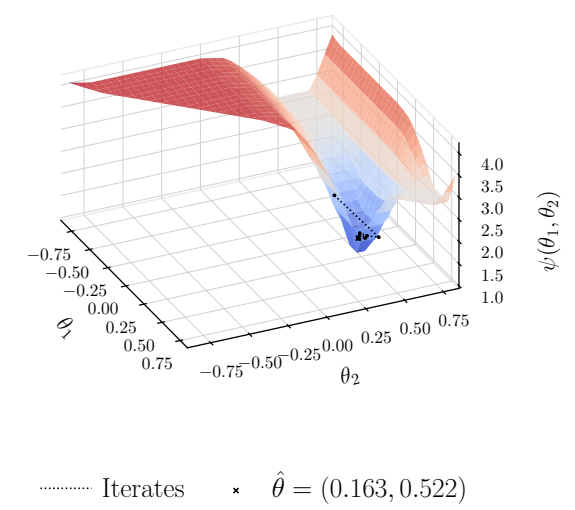
Initial approximation



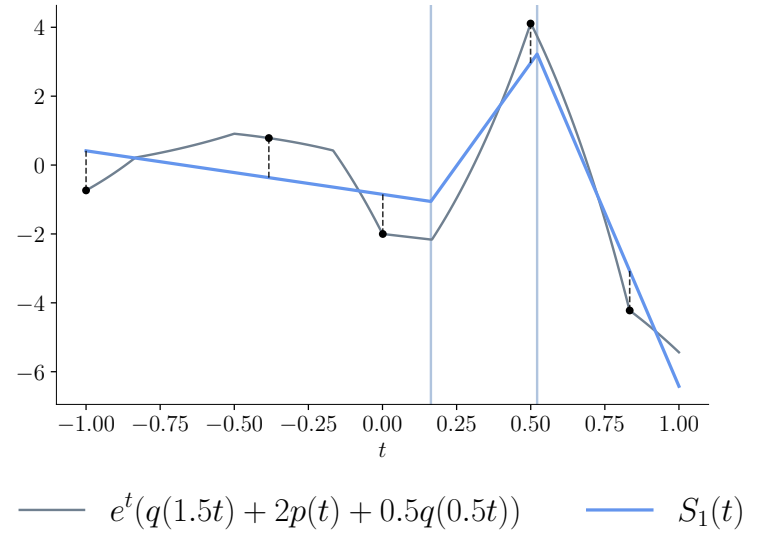
Contour plot and descent path



Objective surface and descent path



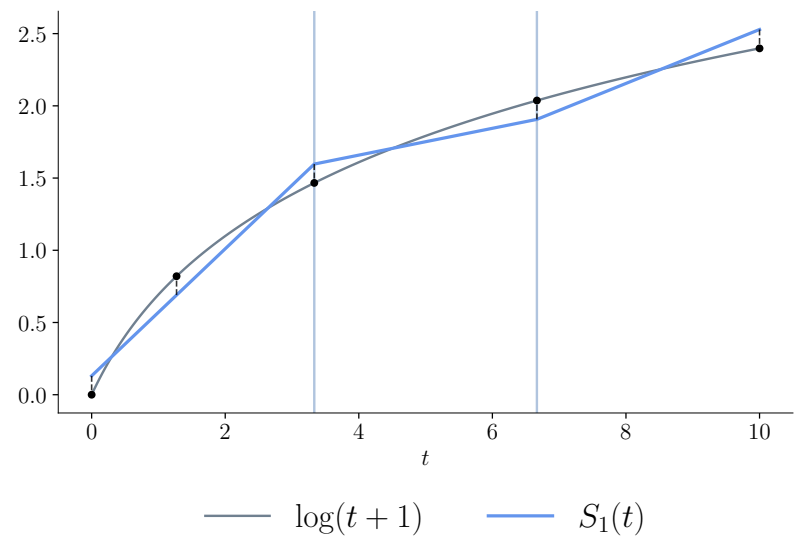
Final approximation



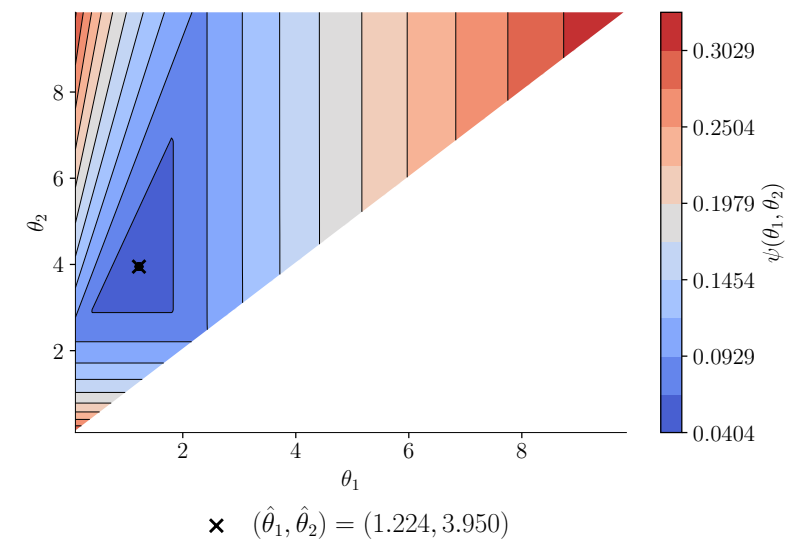
Initial max deviation: 3.12263
 Initial knots: [-0.33333333 0.33333333]
 Nürnberger start: [-0.01226807 0.44944331]
 Theta opt: [0.16329204 0.52158753]
 Final max deviation: -1.15617
 Iterations: 9

log k=2 m=1

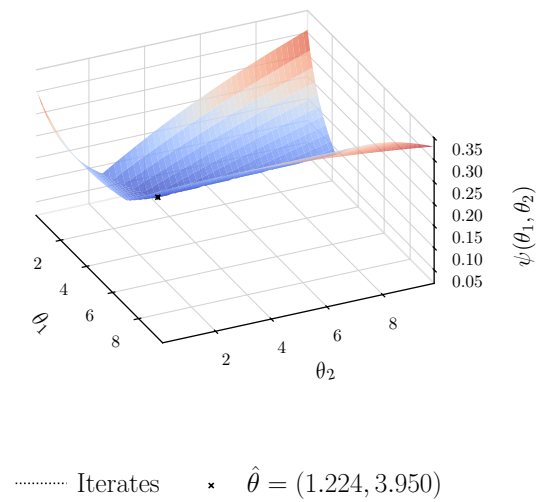
Initial approximation



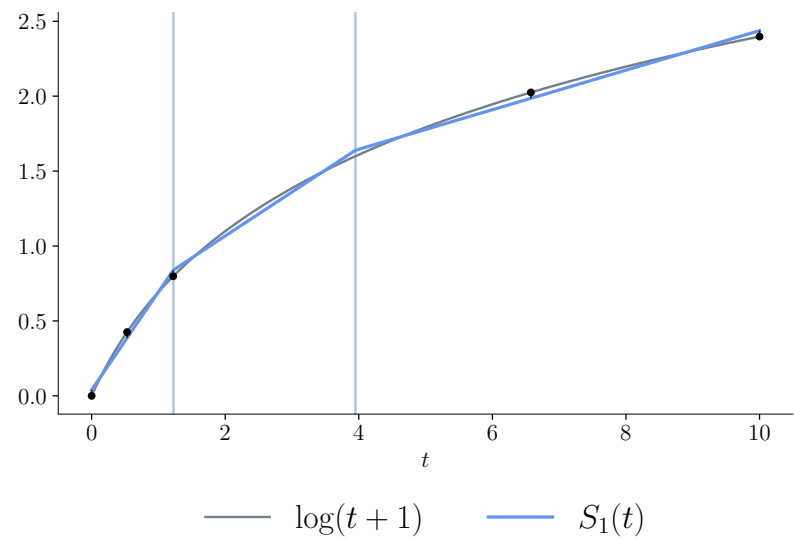
Contour plot and descent path



Objective surface and descent path



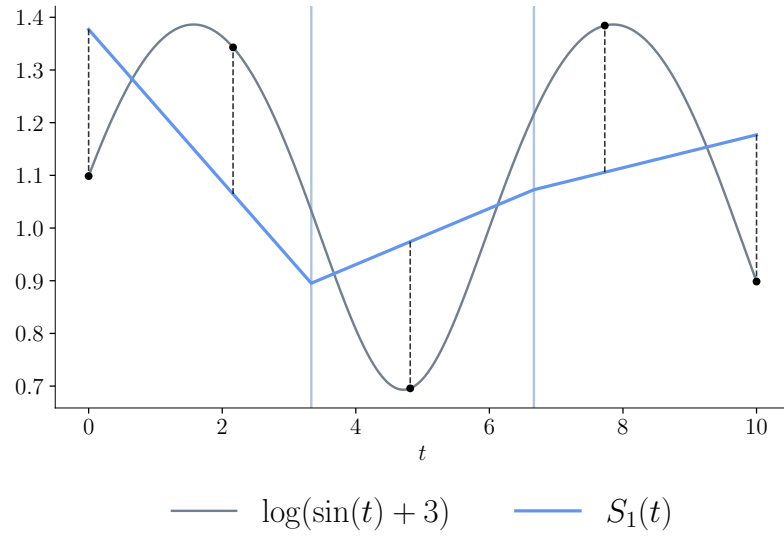
Final approximation



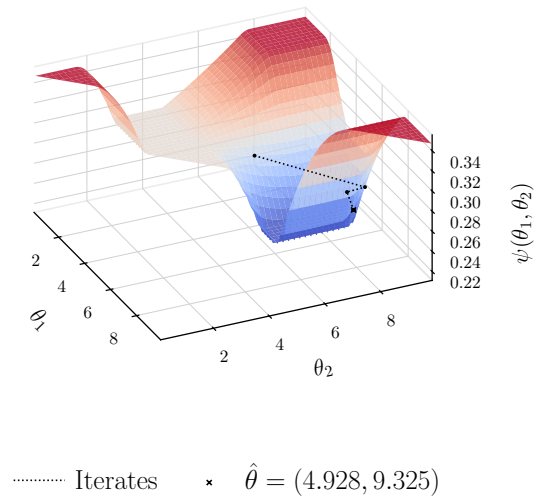
Initial max deviation: 0.130553
Initial knots: [3.33333333 6.66666667]
Nürnberg start: [1.2258594 3.9504528]
Theta opt: [1.22440321 3.94985593]
Final max deviation: -0.0401066
Iterations: 9

log_sin k=2 m=1

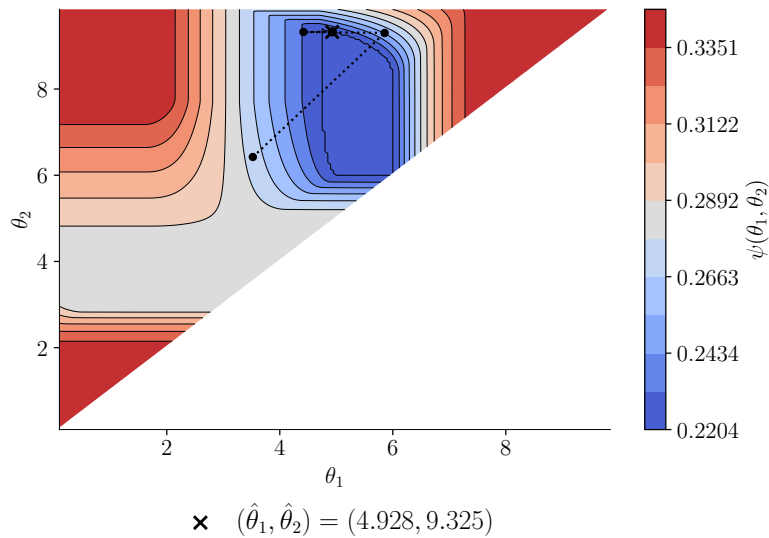
Initial approximation



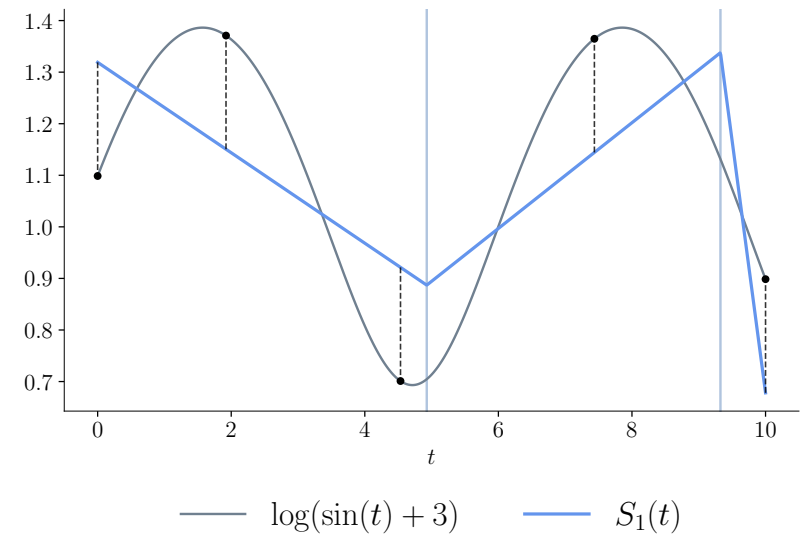
Objective surface and descent path



Contour plot and descent path



Final approximation



Initial max deviation: 0.278388

Initial knots: [3.33333333 6.66666667]

Nürnberg start: [3.52294922 6.42307401]

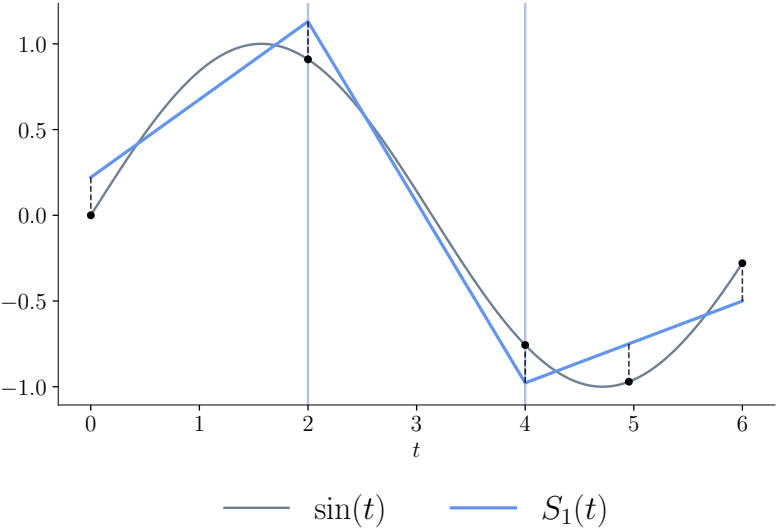
Theta opt: [4.92753421 9.32489774]

Final max deviation: 0.220418

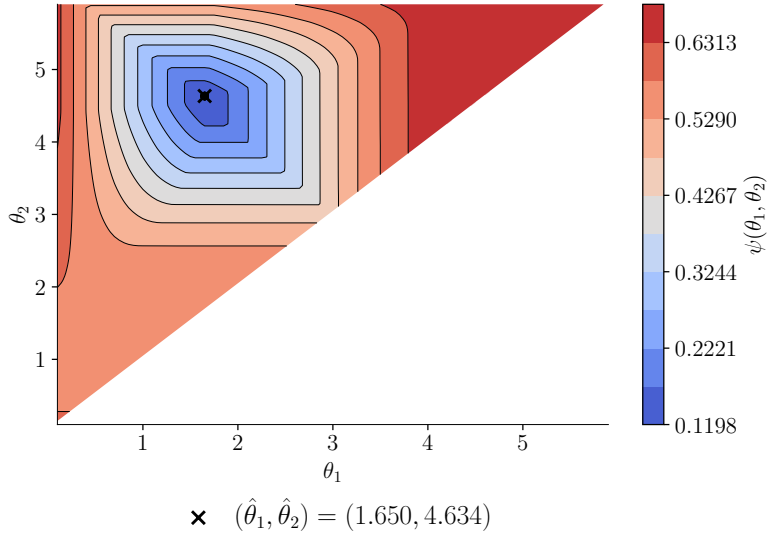
Iterations: 9

sin k=2 m=1

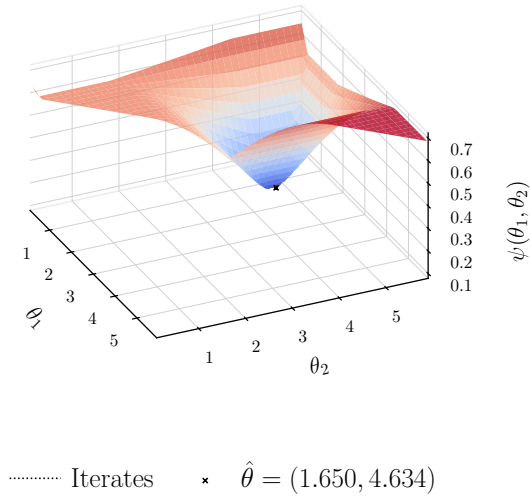
Initial approximation



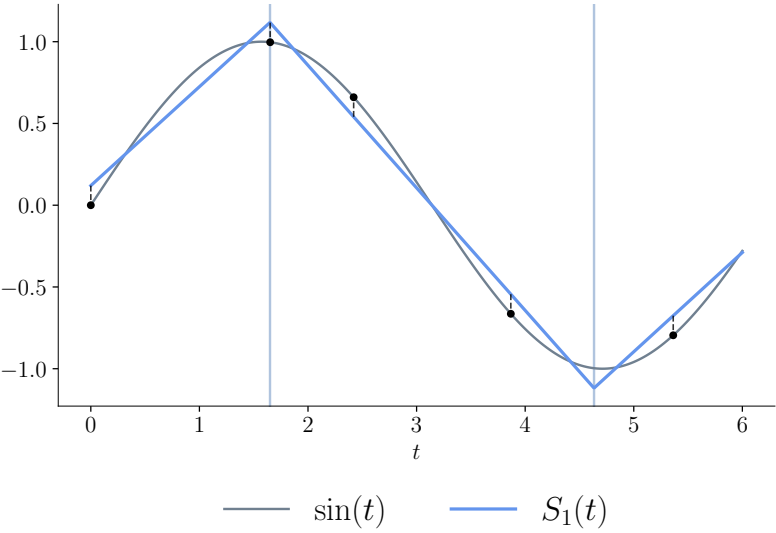
Contour plot and descent path



Objective surface and descent path



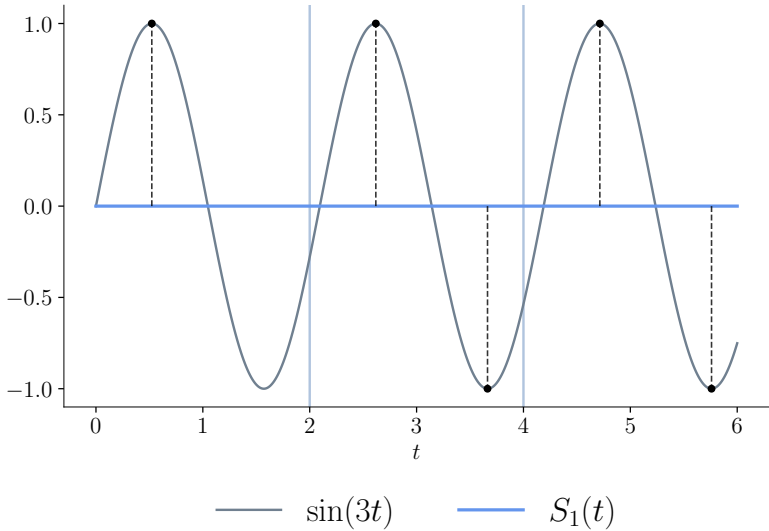
Final approximation



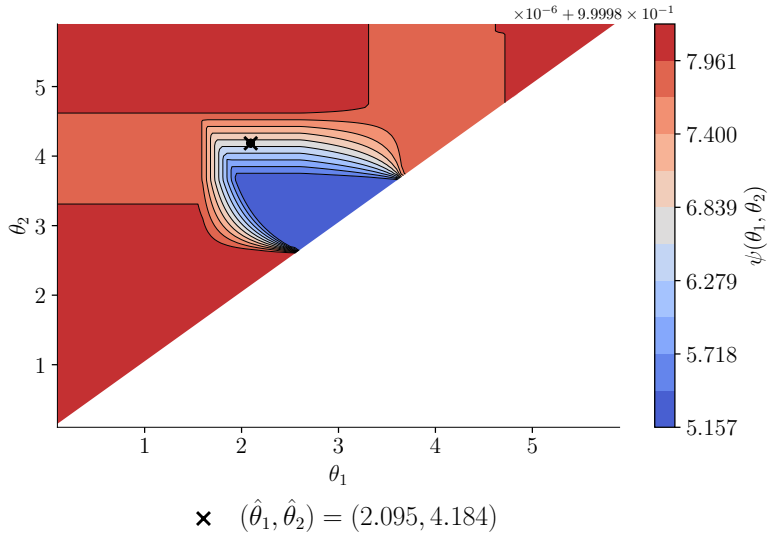
Initial max deviation: 0.220932
Initial knots: [2. 4.]
Nürnberg start: [1.65087891 4.63293618]
Theta opt: [1.65012455 4.63411225]
Final max deviation: 0.121549
Iterations: 9

$\sin 3t \quad k=2 \quad m=1$

Initial approximation

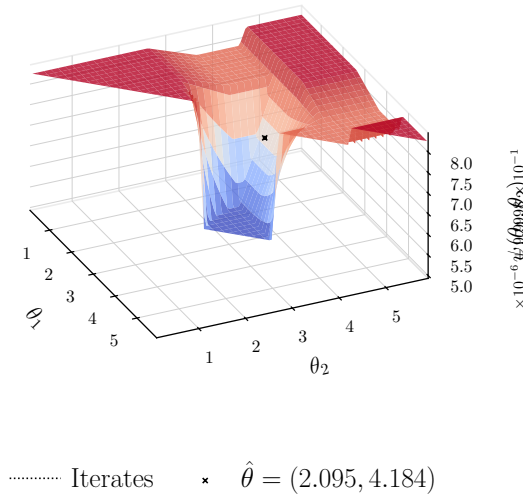


Contour plot and descent path

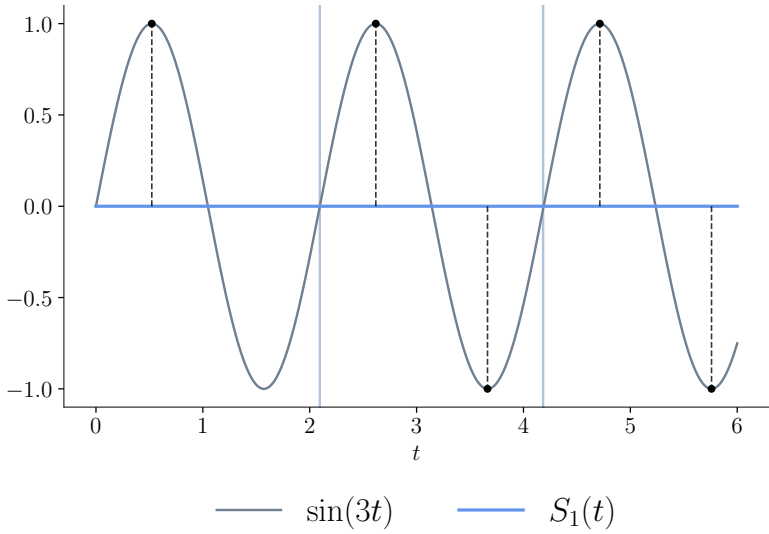


Initial max deviation: 0.999986
 Initial knots: [2. 4.]
 Nürnberg start: [2.09454346 4.1888636]
 Theta opt: [2.09454316 4.18375504]
 Final max deviation: -1.00001
 Iterations: 9

Objective surface and descent path

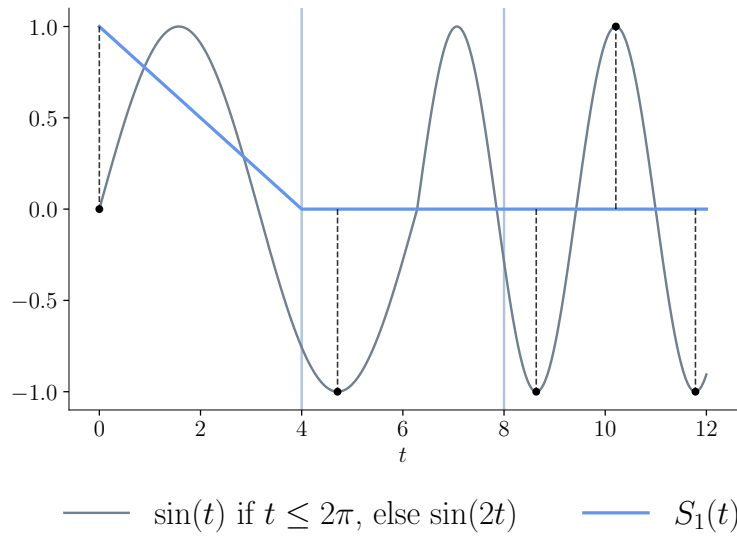


Final approximation

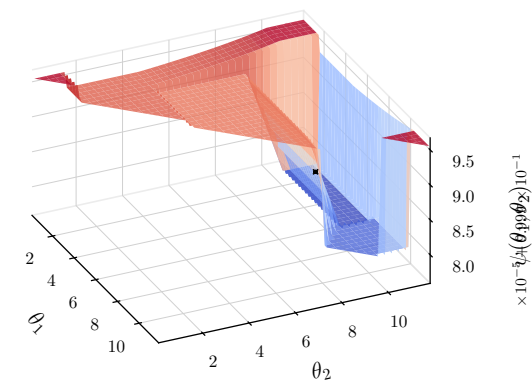


sin_if_else k=2 m=1

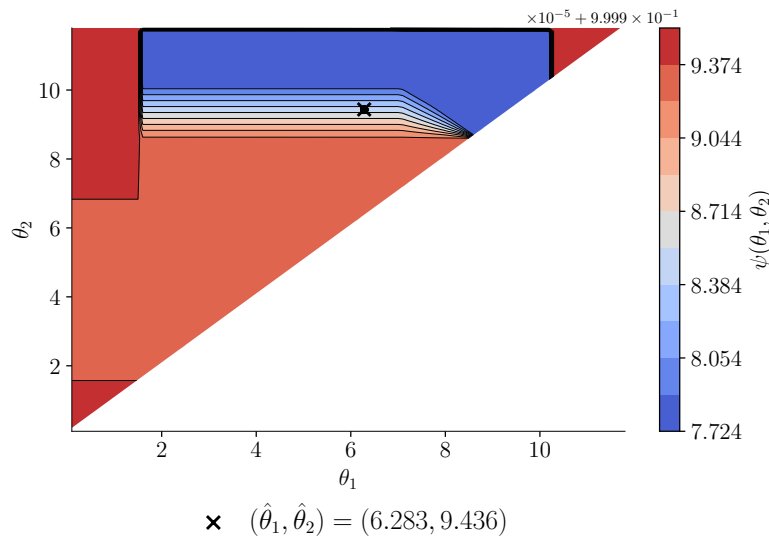
Initial approximation



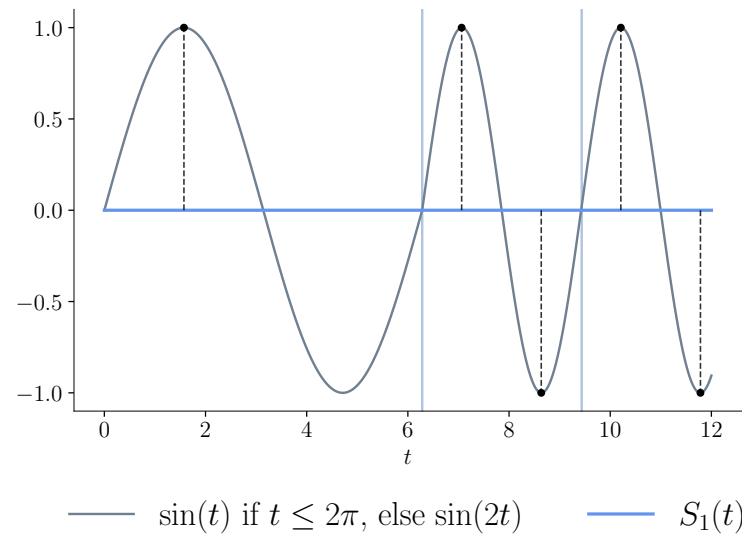
Objective surface and descent path



Contour plot and descent path



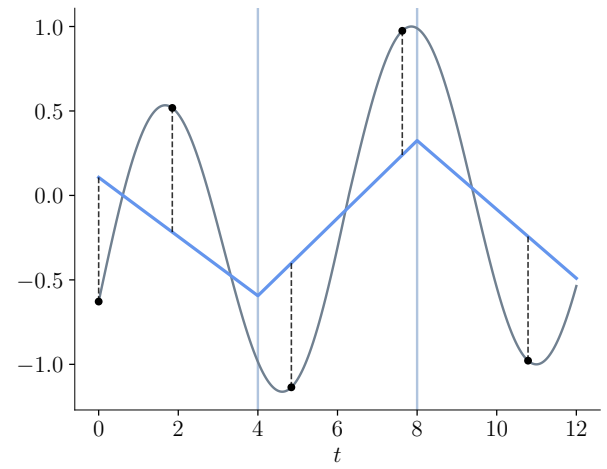
Final approximation



Initial max deviation: 0.999992
Initial knots: [4. 8.]
Nürnberg start: [6.28344727 9.42503911]
Theta opt: [6.28344777 9.43647447]
Final max deviation: 1.00004
Iterations: 9

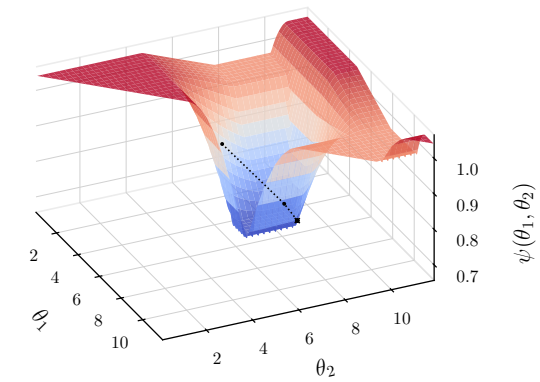
sin_weird k=2 m=1

Initial approximation



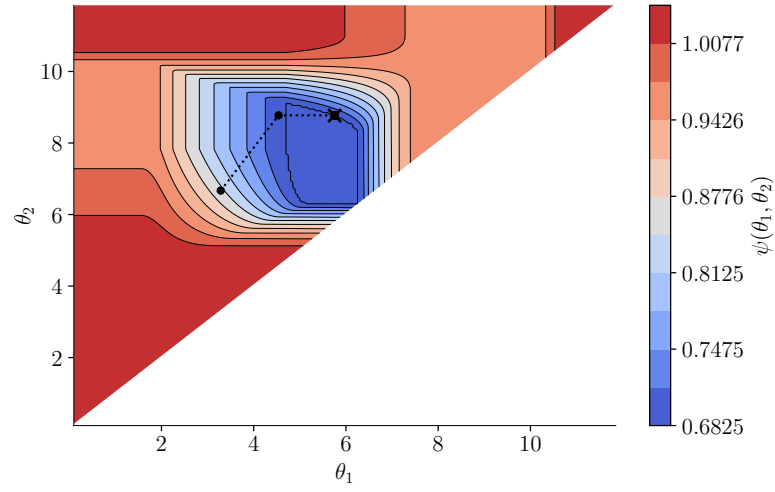
— $\sin(t) + 0.1(t - 2\pi)$ if $t \leq 2\pi$, else $\sin(t)$ — $S_1(t)$

Objective surface and descent path



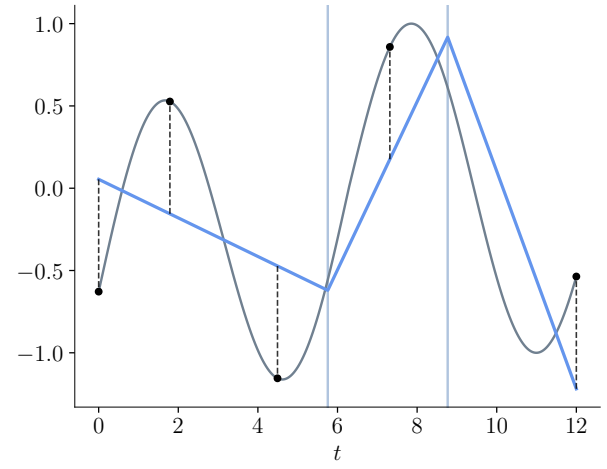
..... Iterates × $\hat{\theta} = (5.756, 8.768)$

Contour plot and descent path



× $(\hat{\theta}_1, \hat{\theta}_2) = (5.756, 8.768)$

Final approximation



— $\sin(t) + 0.1(t - 2\pi)$ if $t \leq 2\pi$, else $\sin(t)$ — $S_1(t)$

Initial max deviation: 0.735199
 Initial knots: [4. 8.]
 Nürnberg start: [3.28564453 6.66947812]
 Theta opt: [5.7559003 8.76800015]
 Final max deviation: 0.682462
 Iterations: 9